

Harshvardhan Singh

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EDUCATION

Rochester Institute of Technology

Master of Science in Computer Science, GPA: 3.53/4.00

Rochester, NY

May 2026

Penn State University

Bachelor of Science in Computer Science, GPA: 3.10/4.00

State College, PA

May 2024

EXPERIENCE

Gen AI & ML Engineering Intern

Big Vision

May 2023 – Jul. 2023

Remote – San Diego, CA

- Full-stack engineer for a GPT-powered document platform architected toward 10,000+ uploads/day: built the React/TypeScript frontend and FastAPI backend linking PaddleOCR extraction to GPT document querying.
- Detection degraded on real scans varying in resolution, orientation, and layout; tuned PaddleOCR hyperparameters and reworked preprocessing to recover quality before documents reached the GPT step.
- Environment drift kept breaking deploys; standardized services under Docker Compose so a full stack came up in one command, and added Pytest/Jest coverage over the upload and extraction paths.

Learning Assistant – Computer Systems

Penn State University

Aug. 2022 – May 2023

State College, PA

- Helped 30+ students debug embedded C and FPGA labs by reproducing their failures, isolating timing and logic issues, and explaining the root cause instead of handing over a fix.
- Automated Gradescope rubric checks so every submission was scored against the same criteria, cutting the hand-grading load for a large lab course.

PROJECTS

CivicPulse | Python, FastAPI, LangChain, React, PostgreSQL, Docker

2026 | [GitHub](#)

- Built a bounded LLM workflow – generate, deterministic claim scan, critique, optional revision – where the scanner outranks the critic: a number absent from the source brief forces a revision anyway.
- Found a bug that attached generated variants to the wrong audience segment while still passing set-level validation; fixed identity resolution against the frozen input snapshot and pinned it with a regression test.
- Classified survey responses with a self-hosted Hugging Face RoBERTa model, keeping model labels and human corrections in separate columns under PostgreSQL check constraints; ~1,100 backend, frontend, and browser tests run in CI.

APIloom | Python, FastAPI, Next.js, PostgreSQL, OpenAI, Docker

2026 | [GitHub](#)

- Built a typed LLM-to-REST planner that turns a prompt into a Pydantic-validated plan, shows the exact resolved request for review, and gates mutations behind tokens bound to plan revision and input digest.
- Hardened outbound calls against DNS-rebinding SSRF: every resolved address is validated, the connection pinned to an approved literal IP through the socket, Host and TLS SNI preserved, redirects re-validated.
- Modeled interrupted mutations with an UNKNOWN_OUTCOME terminal state when delivery cannot be proven, rather than reporting a failure that may have gone through.

Pocket Arcade | TypeScript, Phaser 3, Vite, Vitest, Playwright, Supabase

2026 | [GitHub](#)

- Five responsive browser games with game logic held apart from Phaser by ESLint and a custom import-boundary check, fixed-step updates for frame-rate independence, and seeded runs tests reproduce exactly.
- Wrote 154 unit and 103 Playwright tests, including pixel-signature canvas checks that catch stacked-scene rendering bugs DOM assertions miss; the server re-validates names and score plausibility on submit.

Real-Time Exercise Form Coach | Python, FastAPI, WebSockets, MediaPipe, React

2026 | [GitHub](#)

- Turned browser-side pose landmarks into live rep counts and form feedback using a two-threshold state machine with hysteresis, EMA angle smoothing, and per-joint confidence gates loosened at the knees.
- Designed WebSocket back-pressure that applies control messages in order while dropping stale pose frames, with per-client state isolation and deterministic fixtures covering reconnect and camera-permission paths.

TECHNICAL SKILLS

Languages:	Python, TypeScript/JavaScript, Java, C/C++, SQL, Bash
AI/ML:	LangChain, OpenAI API, Hugging Face Transformers, PyTorch, MediaPipe
Backend & Data:	FastAPI, Node.js, PostgreSQL, Redis, REST APIs, WebSockets
Frontend & DevOps:	React, Next.js, Docker, GitHub Actions
Testing:	Pytest, Jest, Vitest, Playwright